Week 5

In $\mathbb{R}^n$, the mean distance from the start of a random walk is $s\sqrt{t}$ for t number of steps of size s. This is the Root Mean Square (RMS).

Series by Complexity Explorer at Santa Fe Institute.

1. Brownian Motion: Natural world motion and collision of particles in a stochastic manner (observed to be random, but not predictable). This includes stock market fluctuations, diffusion of food coloring in water, cultural diffusion, etc. Random Walk Model is as simple as possible, disregarding collisions. Instead, random directions occur per step.

2. Types of Random Walks:
   - Pearson Random Walk: fixed step, random direction
   - Lattice Random Walk: fixed step, random direction
   - Lévy Flight: power-law step-distribution $\frac{1}{s^{1+\alpha}}$, random direction
   - Shrinking Steps: nth step of length λ^n for $0 < \lambda < 1$. At $\lambda = 1-\varepsilon$, this generates a self-replicating fractal.
   - Growing Steps (Turbulence): diffusion in a random convection field

   Overall:
   - lattice
   - continuous-space short jumps, discrete time
   - discrete-space short jumps, continuous time
   - continuous space and time (diffusion)

   } Central Limit Theorem Domain

   - long-range jumps (Lévy Flights)
   - non-stationary steps (growing/shrinking steps)

3. Root Mean Square Displacement X_{rms}:
   - Consider fixed-step, random direction:  , $|\vec{x}_i| = a$ fixed-step length,
     $\langle \vec{x}_i \rangle = 0$ no bias, $\langle \vec{x}_i \cdot \vec{x}_j \rangle = 0$ for $i \neq j$ is no correlation
   - $\langle \vec{X}_N \rangle = \left\langle \sum_{i=1}^{N} \vec{x}_i \right\rangle = 0$
   - $\langle (\vec{X}_N)^2 \rangle = \left\langle \left(\sum_{j=1}^{N} \vec{x}_i \right)^2 \right\rangle = \left\langle \sum_{i=1}^{N} \vec{x}_i^{\,2} + \sum_{i \neq j} \vec{x}_i \cdot \vec{x}_j \right\rangle = Na^2$ mean square displacement
     (diagonal terms; cross-terms $\to 0$)
   - $X_{rms} = \sqrt{\langle (\vec{X}_N)^2 \rangle} = \sqrt{N}\, a$ for N steps of length a in any dimension
   - $O(\sqrt{n})$ indicates not efficient for exploring space

4. Probability Distribution:

- $P(x,t)$ = probability a random walk is at x for time t
- Consider



- Master Equation: how the probability distribution evolves across time
- $P(x,t) = \frac{1}{2}P(x-1,t-1) + \frac{1}{2}P(x+1,t+1)$ in this example
- $P(r,t)$, r = steps right, l = steps left, $r+l=t$, $r-l=x$
- $P(r,t) = \frac{t!}{r!\,l!}\left(\frac{1}{2}\right)^t$ which is probability for walk right after t steps
- $r = \frac{t+x}{2}$, $l = \frac{t-x}{2} \Rightarrow P(x,t) = \frac{t!}{\left(\frac{t+x}{2}\right)!\left(\frac{t-x}{2}\right)!}\left(\frac{1}{2}\right)^t$
- In general, $\lim_{t\to\infty}$ and by Sterling's approximations, $P(x,t) = \sqrt{\frac{2}{\pi t}}\,e^{-2x^2/t}$ Gaussian Distributions
- $P(x,t+dt) = \frac{1}{2}P(x-dx,t) + \frac{1}{2}P(x+dx,t)$

$$= P(x,t) + dt\frac{\partial P}{\partial t} + \ldots = \frac{1}{2}\left[P(x,t) - dx\frac{\partial P}{\partial x} + \frac{1}{2}dx^2\frac{\partial^2 P}{\partial x^2} + \ldots\right]$$
$$\frac{1}{2}\left[P(x,t) + dx\frac{\partial P}{\partial x} + \frac{1}{2}dx^2 + \ldots\right]$$

- As $dt\to 0$, $dx\to 0$, then $\frac{\partial P}{\partial t} = \frac{(dx)^2}{2\,dt}\frac{\partial^2 P}{\partial x^2}$, $\frac{dx^2}{2dt}$ = diffusion coefficient D.
- Diffusion Equation: $\frac{\partial P}{\partial t} = D\nabla^2 P$ in general dimensions
- $[D] = \frac{\lambda^2}{t}$, $[x^2] = [Dt]$
- $\frac{\partial P}{\partial t} = D\frac{\partial^2 P}{\partial x^2}$. $P(x,t=0) = \delta(x)$. Recall Fourier, Laplace, and Inverses.
- Take $F(k,t)$ of $\frac{\partial P}{\partial t} = D\frac{\partial^2 P}{\partial x^2}$, then take Laplace and inverse, and we get a Gaussian.
- Gaussian we got in general: $P(\vec{r},t) = \frac{1}{(4\pi Dt)^{d/2}}\,e^{-r^2/4Dt}$ for d dimensions

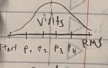
5. Central Limit Theorem

- Walks x, probability $\rho(x)$, assume x are i.i.d variables and $\langle x\rangle$ is finite, and $\langle x^2\rangle$ is finite. Then $P_N(X) = \frac{1}{\sqrt{2\pi N\sigma^2}}\,e^{-(X-N\langle x\rangle)^2/2N\sigma^2}$, $\sigma^2 = \langle x^2\rangle - \langle x\rangle^2$.
- This determines the long-term behavior of the random walk based on first and second moments.
- $\langle X\rangle = \int X P_N(X)\,dX = N\langle x\rangle$, $\langle x^2\rangle - \langle X\rangle^2 = \int x^2 P_N(X)\,dX - \langle X\rangle^2 = N\sigma^2$.
- Fails with step distribution of power-law form.

Code Findings:

Lattice Random Walk

- No matter the sample size, every location has an equal chance of being a starting position. This means all possible areas of the probability distribution have equal consideration. Further, each random direction gives all nearby points equal consideration. Essentially, every repeated trial for n samples will be bias free. The best samples taken from the history based strictly on highest p value discovered means all points in history were considered of equal consideration. This means the selection of the best samples is entirely bias free for the generation of the list to be considered.
- An $m \times m \times \ldots \times m$ lattice is generated with this code. All of $\mathbb{R}^m$ is considered, but this subdivided "m-cube" lattice is how points are selected. This means discrete steps are taken over a continuous normal distribution, and that Lattice Random Walks are subject to the Central Limit Theorem since this current program is completely bias free (except for pseudo-randomness involved). This means, according to the Central Limit Theorem, that the distribution of distances traveled (distance = sum of unique points discovered) will approximate a normal distribution.
- So, the RMS is (step size)·(# of steps), and over that RMS region, the behavior of each random walk from the start is expected to generate a normal distribution of all points discovered between the starting point and RMS. Or rather, the number of times all points are visited will form a bell curve with mean RMS/2, and start and RMS are the bounds of the distribution.



- When bias in random direction probabilities is considered, this distance distribution will change. Since each random step would have a dynamic probability set to pick from (changing after each step), then each step would not be drawn from the same distribution independently. This means a dynamic probability set for step direction would not comply with criteria for the Central Limit Theorem.
- However, if after a trial some fixed probability set for step direction was altered based on the previous trials best steps, then higher probability zones of a non-symmetric probability distribution would be discovered more frequently. And, the generated history would be subject to the Central Limit Theorem.

- For a fixed probability set for step direction, the generated history would be subject to the Central Limit Theorem. If the probability distribution is symmetric, then all directions will tend to the same probability, making such a strategy not useful. If the probability distribution is not symmetric, as most distributions would be, then the generated direction history would reflect the overall direction (or force) to higher probability regions. Since this method would be subject to CLT, then a normal distribution of point visits would be the tendency of the random walk behavior.
- Even if such a normal distribution of local point visits were skewed, the fact that CLT applies means the generated fixed probability set after many trials would be representative of the expected direction of higher probability regions from any starting position.
- This strategy could be used prior to any subsequent strategy as a way to simply gather this expected direction information, as such information would be bias free and follow CLT, which is nice and predictable behavior that is well supported in statistics.
- Computation cost for running the simulation is faster than $O(n^3)$, and fast for fast for large n, t, s. The problem is the visual component of the code with matplotlib. Generating the X_1, X_2 subdivisions and mesh is relatively easy on computation, but something in there is on at least $O(n^2)$.
- For more complex strategies and more data, this computation speed can only slow down. This indicates that simulating with only n samples as provided is logical and computationally reasonable.